
UNIT 16 EMERGING DATABASE MODELS

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16.0 INTRODUCTION

With the advent of relational database systems in the 1970s, database technology became popular in the industry due to the simplicity of database technologies and the availability of SQL for querying the database. However, just a relational model was not sufficient. Many advanced database technologies have become available. We have already discussed some of these technologies, like object-oriented database management systems, data warehousing and mining and NoSQL databases, in the first three Units of this Block. This Unit discusses several advanced database technologies.

This Unit first introduces you to the distributed database systems needed to address the needs of organisations that have distributed data. A distributed database system allows the distribution of fragments of data over a number of database sites. It also supports query processing, which may involve several sites. These concepts are detailed in this Unit. The Unit also introduces you to the concepts of Active databases, XML data and Blockchain technology. Finally, the Unit defines how a database system can be used as a backend to a web application.

This Unit gives a brief introduction to these database technologies. You may refer to the further readings for more details on these technologies.

16.1 OBJECTIVES

After going through this Unit, you should be able to:

- define the need for a distributed database system;
- explain the data fragmentation and replication in distributed databases;
- define the distributed query processing;
- define the features of the active database management system;
- explain the document creation using XML;
- explain the characteristics of Blockchain systems;
- list the characteristics of multimedia database
- use the web database in a web application.

16.2 DISTRIBUTED DATABASES

Many commercial organisations use a database system to manage large amounts of transactional data. These database systems have multiple users and run on a high-performance centralised computer system. These systems may allow geographically distributed users to connect to the database using a network. However, with the increase in the volume of database transactions and user interactions, use of a number of database servers, which may process local data, may be more efficient. This led to creation of distributed database management systems. A distributed database management system (DDBMS) manages several database servers, which may be dispersed at various geographical locations but store the data of a single database system. For example, consider a sample student relational schema given in Figure 1; how can this schema be represented in a distributed database? Figure 2 shows an example of a distribution of the student database over a few possible locations.

Student							
Enrollment No.	Name	Father's Name	Highest Qualification	Email	Phone	Regional Centre	Programme

Fee			
Enrollment No.	Semester	Amount Paid	Date of Payment

Subject		
Enrollment No.	Semester	Subject Code

Figure 1: A sample student relation

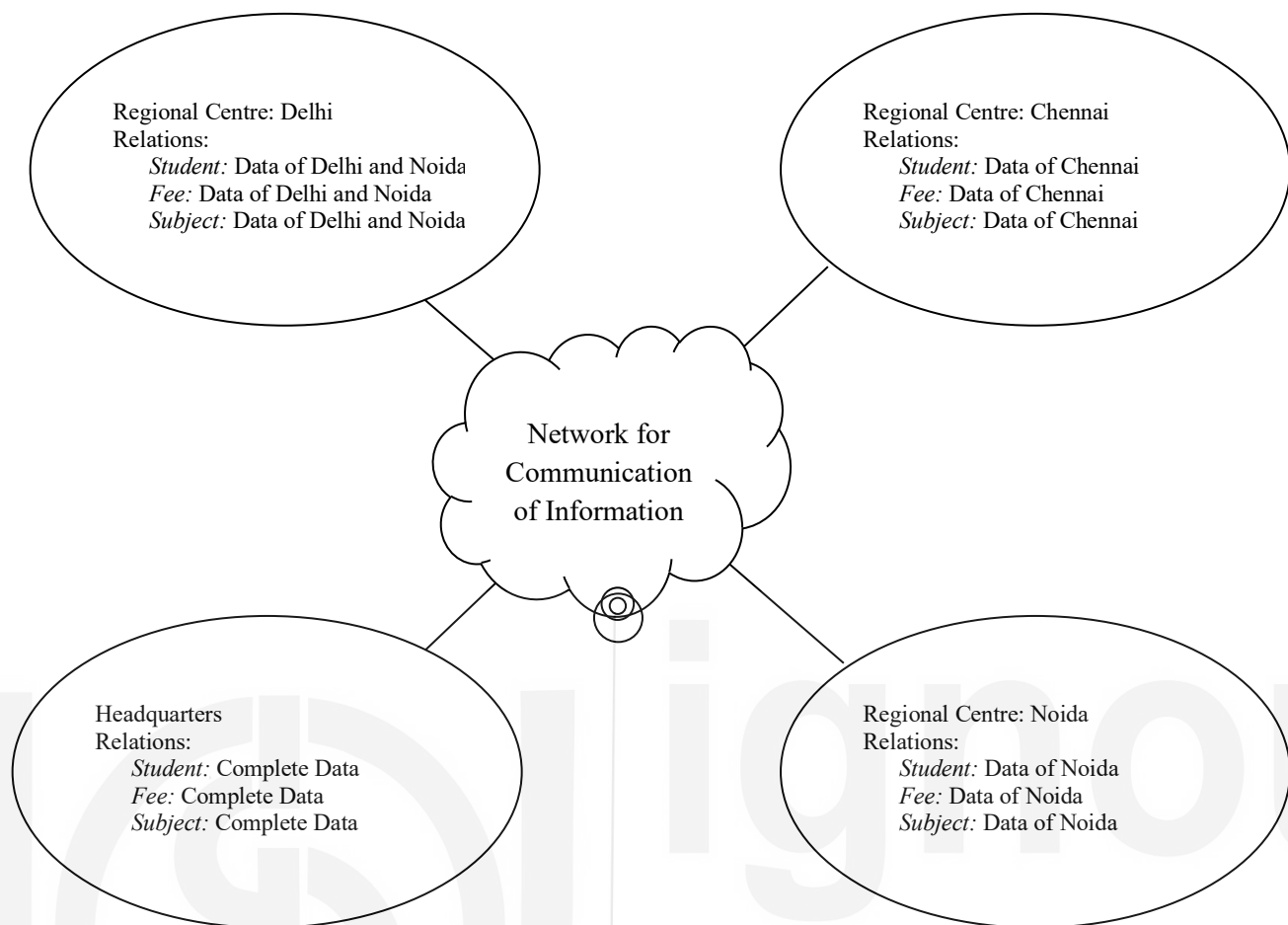


Figure 2: A distributed student relation

The following are some of the important aspects of distributed database systems:

- A distributed database has multiple sites that hold a part of a logically connected database. For example, in Figure 2, the Regional Centre Delhi holds the data of the students who are registered with that regional office of the head office.
- The technical implementation details, such as "The data of Delhi students is available at the Headquarters and Delhi Regional Center", are hidden from the actual users of the database, who can be the students or Regional Centre staff, etc. This is known as transparency.
- The user can issue a command to the DDBMS without worrying about the location of data (called location transparency), where the copies of the data have been kept (called replication transparency), or the fragment of the horizontal fragment of data is kept (called sharding). For example, in Figure 2, a student who is associated with Regional Centre Noida can place his/her query without knowing the fact that his/her data is located at the Headquarters, Regional Centre Delhi and Regional Centre Noida. His/her query will be replied to by any of the locations based on the place of the query and the status of different sites. You may also observe that in Figure 2, the data of students related to Regional Centre Noida is replicated at three sites – Headquarters, Regional Centre Delhi and Regional Centre Noida. Also, note that horizontal fragments of student

data of Chennai are stored at the Regional Centre Chennai site.

- A DDBMS will be required to manage more failures than a centralised DBMS, as it must also manage network failures.
- A DDBMS is available for a longer duration than that of centralised RDBMS.
- A DDBMS is better scalable than a centralised RDBMS due to data distribution on various sites.

16.2.1 Data Fragmentation and Replication

In a distributed database, as shown in Figure 2, all the data is not stored at all the database sites. In general, the data related to a particular site is stored on that site. For example, in Figure 2, data related to Regional Centre Chennai and Regional Centre Noida is stored at their respective sites. The process of distributing data into different parts is called fragmentation. This kind of distribution of data facilitates faster query processing, as most of the queries at a site can be answered from the local data. In addition to fragmentation, the data is replicated at more than one site. For example, in Figure 2, data of Regional Centre Noida is replicated at Regional Centre Noida and Regional Centre Delhi sites. This data replication helps improve the reliability and availability of the database, as the database would be available to users even if one of the replicated sites fails. The following example explains different kinds of fragmentation.

Example: Consider the slightly modified Student table given in Figure 1 with the following database instance:

Student						
EnrNo	Name	Father	Qual	Phone	RC	Programme
23001	Anil	Mohan	UG	9900100000	Noida	PGDCA
23002	Rahim	Jamil	UG	9900200000	Chennai	PGDCA
23003	Simon	Robert	PG	9900300000	Noida	MCA
23004	Sahil	Sanjay	UG	9900400000	Delhi	MCA
23005	Sanjay	Ajay	PG	9900300000	Noida	PGDCA
23006	Diya	Jeba	UG	9900400000	Chennai	MCA

The following can be the Horizontal Fragments of the table based on the RC.

Student: Horizontal Fragment on RC Noida site						
EnrNo	Name	Father	Qual	Phone	RC	Programme
23001	Anil	Mohan	UG	9900100000	Noida	PGDCA
23003	Simon	Robert	PG	9900300000	Noida	MCA
23005	Sanjay	Ajay	PG	9900300000	Noida	PGDCA

Student: Horizontal Fragment on RC Chennai site						
EnrNo	Name	Father	Qual	Phone	RC	Programme
23002	Rahim	Jamil	UG	9900200000	Chennai	PGDCA
23006	Diya	Jeba	UG	9900400000	Chennai	MCA

Student: Horizontal Fragment on RC Delhi site (Noida or Delhi)						
EnrNo	Name	Father	Qual	Phone	RC	Programme
23001	Anil	Mohan	UG	9900100000	Noida	PGDCA
23003	Simon	Robert	PG	9900300000	Noida	MCA
23004	Sahil	Sanjay	UG	9900400000	Delhi	MCA
23005	Sanjay	Ajay	PG	9900300000	Noida	PGDCA

The other kind of fragmentation is vertical fragmentation. For example, if RC Noida has been assigned the work of contacting all the students telephonically, irrespective of their regional centre, then one possible vertical fragment for Regional Centre Noida would be as follows:

Student: Vertical Fragmentation for Noida			
EnrNo	Name	Phone	Programme
23001	Anil	9900100000	PGDCA
23002	Rahim	9900200000	PGDCA
23003	Simon	9900300000	MCA
23004	Sahil	9900400000	MCA
23005	Sanjay	9900300000	PGDCA
23006	Diya	9900400000	MCA

Further, considering that Regional Centre Noida is assigned PGDCA students only for telephonic contact, the fragment would be a mixed fragment as follows:

Student: Mixed Fragmentation for Noida			
EnrNo	Name	Phone	Programme
23001	Anil	9900100000	PGDCA
23002	Rahim	9900200000	PGDCA
23005	Sanjay	9900300000	PGDCA

As far as replication is concerned, you may observe that the student data of Regional Centre Noida is replicated at Regional Centre Delhi, too. Replication helps in enhancing reliability and availability but results in more overheads in transaction processing.

16.2.2 Distributed Query Processing

A query in a distributed database management system is submitted at a site. This query is then converted to a relational algebraic query and optimised using local and global query optimisation processes. Local query optimisation is the same as that of a centralised DBMS; however, global query optimisation involves the selection of sites for query evaluation, cost of data communication, and cost of query processing. The process of distributed query processing is explained with the help of the following example.

Example: Consider a query submitted at the Headquarters seeking to find the Percentage of fee share of Regional Centre Noida in the financial year 2022-23. This query would require computing the fee collected by RC Noida to the total fee collected between the dates 01st April 2022 and 31st March 2023. This query may consist of two subqueries:

(a) Finding the total fee collected for the financial year 2022-23.

(b) Finding the total fee collected at RC Noida in the financial year 2022-23.

In addition, you may assume that the financial records are ordered chronologically.

One of the possible ways of processing the queries would be to process both the sub-queries at the Headquarters. If both these sub-queries can be processed during the same query processing cycle, then it may be a good choice. However, if both the sub-queries are processed separately, then other options may be explored. What if subquery (a) is processed at Headquarters and subquery (b) is processed at the RC Noida site? This will require a transfer of results from subquery (b) from the RC Noida site to the headquarters site, where the result will be displayed to the user who made the query.

A detailed discussion on distributed database management systems is beyond the scope of this Unit. You may refer to the further readings for more details on this topic.

16.3 ACTIVE DATABASES

Active databases, as the name suggests, comprise dynamic actions on the occurrence of certain events. Such actions were part of the SQL 99 standard and are called triggers. Let us define the model that can be used for an active database:

Active Database Model

Consider the Student and Result relations given in Figure 3.

Student				
Enrollment No.	Name	Programme	Cumulative Grade Point Average	Status

Result		
Enrollment No.	CourseCode	Grade

Figure 3: An example

Assuming that the Cumulative Grade Point Average is to be updated for each student when related data is created in the Result relation. The following events will cause a database action to be activated:

Action Triggering Event: In the database of Figure 3, on updating the Result table, the Cumulative Grade Point Average (CGPA) of a Student needs to be updated, as well. Assuming that once a Record is entered in the Result table, then it cannot be deleted, and only the Grade attribute can be modified in the Result relation, the following may be the events that may trigger the action of an update on CGPA for the Student relation:

- Addition of a tuple in the Result relation
- Modification of a Grade in the Result Relation.

Once the trigger event occurs, the next step is to check the condition, if any. In the case of the Student database given in Figure 3, the referential integrity constraints on the Enrollment number and CourseCode (each student will register for a set of courses) will make sure that only the valid entry is made in the Result table. In addition, the check constraint on Grade ensures that a valid Grade letter is entered in the Result table. However, an interesting observation here is that the state of a student must be "Active" in case his/her CGPA is to be updated. This can fit as a condition for activating the trigger.

Issues relating to Active databases: Some of the issues relating to active databases relate primarily to the process of creating and maintaining the triggers. Such databases must include a set of rules or commands that should be able to deactivate or drop the old or redundant rules, as there can be a very large number of rules which may be changed during the lifetime of the database. Next, since triggers can be activated either before, during or after a condition, a huge number of events may be scheduled, causing the database to act slowly as a response to normal record addition and deletion processes. Interestingly, it adds another issue, considering that addition of results by a teacher in the student's database triggered the update of about 50 students' GPA. However, the teacher realised that s/he had used an incorrect file to update student records and deleted his/her result. This will lead to the firing of a very large number of triggers that would be required to undo the changes in the GPA of the students.

A detailed discussion on active databases is beyond the scope of this Unit. You may refer to further readings for more details.

Check Your Progress 1

1) How is DDBMS different from RDBMS?

.....

- 2) Suggest the data fragmentation of the student table if all RCs need only the enrolment number and name of all the PGDCA students.

.....
.....
.....

- 3) What is an active database? What is a triggering event?

.....
.....
.....

16.4 XML AND DATA REPRESENTATION

The eXtensible Markup Language (XML) is one of the popular data representation languages. It uses user-defined tags to represent a document. A typical XML document relating to the student's table, as shown in Figure 3, is shown below:

```
<school>
  <class>
    <class_no>XII</class_no>
    <class_teacher>John</class_teacher>
    <student>
      <enrolmentNo>2301002297</enrolmentNo>
      <name>Ritesh Jain</name>
      <programme>PGDCA</programme>
      <cgpa>7.5</cgpa>
      <status>Pass</status>
      <result>
        <coursecode>BCS011</coursecode>
        <grade>A</grade>
        <coursecode>BCS013</coursecode>
        <grade>B+</grade>
      </result>
    </student>
  </class>
</school>
```

```
<student>
  <enrolmentNo>2301002301</enrolmentNo>
  <name>Amitesh</name>
  <programme>PGDCA</programme>
  <cgpa>8.5</cgpa>
  <status>Distinction</status>
  <result>
    <coursecode>BCS011</coursecode>
    <grade>A+</grade>
    <coursecode>BCS013</coursecode>
```



```

        <grade>A</grade>
    </result>
</student>
</class>
</school>

```

Figure 4: A sample XML document

You may please observe that instead of using a separate table, in the XML document, the results of the students are merged along with the student information. Thus, XML representation has the potential to store all the information about an entity in one place. Such a representation, though it may be useful for searching from a point of view as no join operation is required, may lead to redundancy of information. In addition to the use of tags, XML allows users to store attributes along with a tag. For example, the enrolment number can be stored as an attribute of the student as:

```

<student enrolmentNo = "2301002301">
    <name> ...
</student>
...

```

The attribute may be useful for searching for information on the related field.

Just like the database management system has a different schema, XML also can be used to validate the structure of the XML data. Figure 5 is a possible document type definition (DTD) that would validate the document given in Figure 4.

```

<!DOCTYPE school [
<!ELEMENT school (class+)>
<!ELEMENT class (class_no, class_teacher, student+)>
<!ELEMENT student (enrolmentNo, name, programme, cgpa, status, result *)>
<!ELEMENT result ( (coursecode, grade)+)>
<!ELEMENT class_no( #PCDATA )>
<!ELEMENT class_teacher( #PCDATA )>
<!ELEMENT enrolmentno( #PCDATA )>
<!ELEMENT name ( #PCDATA )>
<!ELEMENT programme ( #PCDATA )>
<!ELEMENT cgpa ( #PCDATA )>
<!ELEMENT status ( #PCDATA )>
<!ELEMENT coursecode ( #PCDATA )>
<!ELEMENT grade ( #PCDATA )>
]>

```

Figure 5: Document Type Definition for XML of Figure 4

Please note the following points in the DTD, as shown in Figure 5:

- A school consists of data from one or more classes.
- Each class stores the class number and class teacher's name.
- A class have one or more students.

- For each student, you store the enrolment number, name, programme, his/her CGPA and present status (Distinction, Merit, Pass, Unsuccessful).
- In addition, the result of each student is stored. This result can be in zero or more subjects.
- The filed type #PCDATA means parsed character data, which stores the text parsed by the parser into the fields.

XML has become a popular format for data exchange and storage. Several tools have been developed to verify, display as tree nodes and query the XML documents. A detailed discussion of these tools is beyond the scope of this Unit. However, we present a few basic features of XQuery, which is a standard query language for XML documents.

A XQuery expression uses five basic keywords for querying. These are *for*, *let*, *where*, *order by* and *return*. The *for* clause selects a sequence of nodes in a document, *let* is used to bind a variable name to a sequence, *where* is used for the selection of nodes, *order by* is used for giving an order to the sequence, and *return* is used to specify what values are to be returned.

For example, the query:

```
for $x in /school/class/student
where $x/cgpa > 8
return </name>
```

This query will return the name of the student with a CGPA of 8 or more. You may refer to the further readings for more details on XQuery.

16.5 BLOCKCHAIN DATABASES

Conceptually, blockchain is a paradigm of storage of data in a distributed manner, which may also protect data from fraudulent transactions and updates. Blockchain technology was used to store distributed ledgers and bitcoins. However, this technology is not limited to only these applications. In this section, we will present some of the basic features of blockchain with the help of an example. However, to understand the principles of blockchain, you should study the paper "Bitcoin: A Peer-to-Peer Electronic Cash System" by Satoshi Nakamoto in 2008.

Let us discuss the components of the blockchain technology. A Blockchain will have the following components:

1. Distributed set of Authorised Nodes: The role of a node is to keep the ledger of data. A ledger consists of data logs, which are timestamped. A *blockchain* is a sequence of *blocks* of data, which is maintained at each of these authorised nodes.
2. A *block* contains:
 - a. Block Number: It is a unique sequence number given to every

block on the blockchain.

- b. Nonce: It is a random number that is used only once in a block.
 - c. Transaction logs: A sequence of transaction logs that are to be recorded in a block.
 - d. Computed Hash value: A cryptographic hash function is applied to the content of a block to compute a hash value of the block. This hash value is also stored in the block.
 - e. Hash value of Previous Block.
3. A blockchain is constructed by linking the blocks. In addition, each block contains the hash value of previous block, which is used to ensure data security.

A hypothetical blockchain is shown in Figure 6.

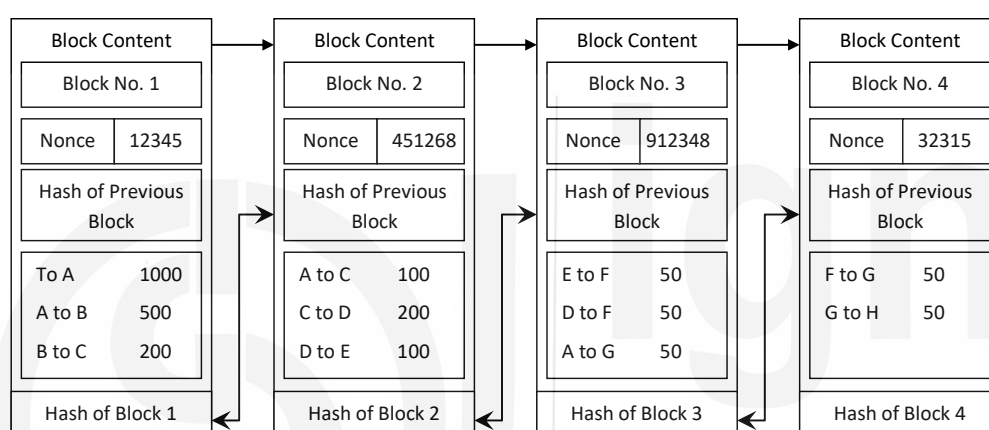


Figure 6: A Hypothetical Blockchain Data

The blockchain processes involve the following:

1. A network consensus protocol, which ensures security, trust, and concurrence amongst the network of nodes of the blockchain.
2. The hashing process ensures the integrity of the transaction ledgers; in other words, it ensures that the transaction ledgers are not tampered with. In Figure 6, each transfer of money may represent one entry of the transaction ledger.
3. Uses a digital signature to certify consent for the transaction.

Let us discuss some of these concepts in more detail.

Cryptographic Hash Function: At the most elementary level, a blockchain consists of a cryptographic Hash function. This hash function is a kind of secure fingerprint of a Block of data of blockchain. One of the popular hash algorithms is Secure Hash Function 256 (SHA256), which produces a 256-bit long hash value for the content of a block irrespective of the size of the block (which can vary from one character to millions of characters). The hash function has the following properties:

- It maps the block contents of any size to a hash value called Hash of a fixed size; for example, SHA 256 maps block content of any size to 256 bits Hash.
- For a given content, the hash function will always produce the same Hash. In addition, it is expected that if the contents of two blocks are different even by a single character, then the hash function should produce different Hash values for the blocks.
- In case you change the content of a block slightly, then the value of the Hash function changes significantly. In addition, if you know a part of the hash value, you cannot predict the rest of the hash value.
- The cryptographic hash function is also called one-way encryption, as you can use the block contents to create the Hash value, but you cannot use the Hash value to find the block contents. Interestingly, many password-based authentication systems store the password using one-way encryption. This is why your system administrator simply provides you a link to change the password, but not the password itself.

Please also note that in case there is a change in the content of a block, its hash value will also change. This feature of the hash value may be very useful in finding the tampering of data of a block, which is discussed next.

How to identify data tampering or corruption in a Blockchain?

As discussed earlier, any minor change of any value in a block at a particular node will result in change in its computed hash value. This will cause change in the stored *hash value of previous block* in all the subsequent blocks. For example, consider in Block No. 2 of Figure 6, the transaction log of A to C is modified to 200 from the present value of 100. This will result in change in the computed hash value of Block no. 2. Further, this will result in change in *hash value of previous block* of Block 3, which will result in change in the computed hash values of Block no 3. A similar change will occur in Block 4, too. Thus, the previously stored hash values of Block 2, Block 3 and Block 4 will not match the computed new hash values of these blocks.

But how will it be recognised that the blocks have different hash values? Here, the consensus protocol will play its role. Since each blockchain block is stored on all the blockchain network nodes, changes in one site can easily be recognised, as the other nodes with the stored copy of the blockchain will not agree with this blockchain, thus identifying the tampering.

Use of Nonce: Nonce is part of the block content and is a short form of random Number used once. This number is mined using an algorithm such that the Hash value generated for the block is unique.

Validity of Transactions: You may observe that in Figure 6, most of the transactions show the transfer of money from one account to another. For example, in Block 1, there is a transaction - A to C 500. But does A have that much amount? Well, that can be established from the first transaction in this block, which says "To A 1000". Thus establishing that A has sufficient funds for

the transfer. Such transactions are sometimes called Coinbase transactions and can be used to ascertain the validity of transactions.

Use of Digital Signature: The digital signatures are used to ensure that the transactions are performed by the authorised person. For example, consider a new transaction, "A to E 50", to be added to Block 4; how will it be ensured that this transaction has been performed by A? For this, A needs to have a pair of private and public keys; let us call them PrivateKeyA and PublicKeyB. Both these keys can be very long in length. A must maintain the PrivateKeyA as a secret key, whereas he can share the PublicKeyA with other stakeholders. One key characteristic of this private-public key pair is that if you know the PublicKeyA, you cannot derive the PrivatekeyA. A can create this transaction "A to E 50" by using the PublickeyA and PublickeyE as:

"PublickeyA toPublickeyE 50"

Next, A needs to sign this message using the PrivatekeyA. This transaction can be verified by using the PublickeyA, which ensures that the transaction has been created by an authorised person only. Please note here that in Figure 6, we have shown transactions using alphabets like A, B, C, etc., but in the actual blockchain, they will be public keys.

You may refer to the further readings for more details on Blockchain technology.

 Check Your Progress 2

- 1) Create an XML document consisting of Marks of two students in at least one subject. Also, make the DTD for validating this XML document.
.....
.....
- 2) What is the role of nonce in a blockchain?
.....
.....
- 3) Consider that in Figure 6, in Block 3, the E to F transaction is modified to 100. What changes would it cause in the blockchain?
.....
.....

16.6 MULTIMEDIA DATABASE

Multimedia data is an integration of textual, graphical, audio, video, and animation data. In general, multimedia data may include lengthy textual documents, pictures, drawings, digital audio clips, movies, and animations. A multimedia database should be able to store large multimedia data efficiently and provide the feature of querying the multimedia data.

Querying is a very interesting domain in the context of multimedia data, as most searches in such data require retrieval of data based on some content. For example, you may be interested in all the videos related to "Database Integrity and Normalization" from a multimedia database or videos of a particular presenter. Please note that such queries would require indexing on the objects and related contents.

How can you create these indexes? One way to create indexes would be to create a large amount of metadata for each object. This metadata should use standardised keywords, title, credentials of creators, summary information, etc. However, in many situations, the metadata would not be available, leading to the generation and verification of the metadata. The second type of indexes are those that can be created using automatic analysis.

Characteristics of Multimedia Data:

Let us now discuss some of the basic characteristics of different types of multimedia data.

Textual Data: In general, the textual data consists of articles, which include paragraphs and headings. This data is stored using ASCII or Unicode standards. Further, you may define certain keywords on these articles that can be used for searching.

Image Data: An image is stored in digital form using picture elements called *pixels*. The size of the picture depends on the number of colours used. For example, a black and white picture would just require 1 bit for every pixel, whereas a true coloured picture may require 24 bits (8 bits each for three basic colours) to store one element. Further, the resolution of a picture is represented using pixels per inch. Therefore, the size of a good-quality picture is quite substantial. You can use different kinds of compression standards to reduce the size of a picture. Some of the common image files include formats like GIF, PNG, JPEG, etc. A search on images may be to find the images related to the same objects. To answer such a query, every image can be divided into segments that have similar characteristics. Further, this segment characteristics information can be grouped to identify certain basic characteristics of an image. Various images are compared based on the similarity of characteristics of those images. In addition, a number of labels, indexes, etc., can also be linked to the segments of images, helping in finding meaningful information about the images. You may have noticed that present image recognition software are able to link many images together based on their characteristics.

Video and Animation Data: Video and animation data may be considered as a large sequence of frames that are displayed in real-time to form a live movie. In general, video data may be divided into video segments, which may be related to a group of objects or activities. For example, an eLearning video may be divided into segments, with each segment involving a sequence of learning concepts. Some of the popular compression formats used for video are MPEG, AVI, etc. A query of video data may be related to identifying the portion of a video related to a specific learning objective.

Audio data: Audio data is recorded audio messages. These messages may be identified for similarity of voice. A query on audio data may try to group the audio of a person based on recognition of voice.

Issues and Characteristics of Analysis of Multimedia Data

- Multimedia data must be stored, labelled, and indexed so you can define some similarity measures on that data.
- Statistics can be used to define the characteristics of image data using the values of colour, texture, etc.
- Further, the object's shape can be one feature of object recognition. This may

include the identification of facial features.

- In the present time, the recognition of an object in an image, video or animation is a major challenge. The key barriers to object recognition are:
 - The angle from which an image has been taken may vary the shape of an object
 - The scale and size of the picture may affect the recognition of objects.
 - Rotated, transformed, and occluded images are difficult to recognise, as it is difficult to keep a database of all transformations of an image.
- An important development in the area of object recognition is the development of a scale-invariant feature transform (SIFT) by David Lowe. SIFT extracts features from images such that they are not affected by rotation and scaling. You may refer to this algorithm in further readings.
- Another important concept relating to the recognition of images is the tagging of images. Tagging is useful in searching for images. You must have observed on certain social media that you may be tagged in some images that include you. At present, the tagging algorithms are being enhanced for better accuracy. In general, these algorithms use machine learning and statistics to analyse the image content with already tagged image libraries.
- Digital audio is difficult to index and retrieve, as it does not have any typical features or characteristics that can be used to identify the content specifically.
- A multimedia database manages different data formats and, thus, requires many storage and Input/Output technologies. These technologies may include technologies for text and images, like scanners, digital cameras, printers, etc.; technologies for audio, video, and animation, like microphones, video cameras, DVDs, Musical Instrument Digital Interface (MIDI), good quality displays and speakers etc.

Multimedia data is part of many digital systems, such as Patient health monitoring data, geographical data, students' data, etc.; therefore, effective multimedia data management systems are required to handle such vast data.

16.7 USE OF DATABASES IN WEB APPLICATIONS

A database system is a persistent collection of an organisation's data, which is shared and integrated among various applications. Database technology supports non-redundant storage of data, which allows the following features:

- Structured storage of data in the form of tables
- Secure data insertion, modification, and deletion.
- Support for concurrent database transactions.
- Easy but controlled access to data.

These features are very useful for any web application, too. Therefore, many web applications use database management systems (DBMS) to store data and access it securely for display on the web. These DBMSs are normally managed on a separate server, called a database server, and communicate with the web server to store or retrieve data. The web server then communicates this information through the relevant web pages to communicate with the web clients.

For example, when you register on an eCommerce website, the information you fill in the registration form is stored in a registration database. This registration information is accessed when you log in again on that website. On the eCommerce website, you may

also search for product information, put some items in the shopping cart, order the items from the cart, and so on. All these activities are supported by several databases. However, these activities are obscure, as you just interact with the eCommerce website through a browser interface. This section focuses on how the database can be accessed by a web server; however, our approach will be to discuss the process rather than the use of a scripting language for coding.

Some of the key characteristics that should be supported by the web server–database server interconnection are:

- The information exchange between the two servers must be secure.
- The two servers should have pools of connection for transfer of information.
- Concurrent read and write operations should be supported by the database server.
- The response time for an operation should be as low as possible.

In general, these servers may either be supporting a 2-tier or a 3-tier client-server architecture.

Steps for Creating a Web Database Application

The first requirement for creating a web application is that your web server has the required drivers to connect to the DBMS of interest. You are required to perform the following steps to create a web application that uses a database as a backend.

Connecting Web Server to a Database Server

On receiving a request from a web application client, a web application may request data from a database server. In general, a web server is required to establish a connection with the database server if it has not already done so. You require the following parameters for a connection:

- A database driver for the DBMS of interest.
- Access credentials on the DBMS and the database (e.g. username and password).
- The URL of the database server and its port number.

For example, assume that you are developing a web application using Java Server Pages (JSP) and you want to connect to a MySQL database. Further, your username is *student* and password *dbms*. In addition, the MySQL host is a local host at the URL and port number - *jdbc:mysql://localhost* and 3306, respectively. Further, assume that the database which is to be accessed is named: *class_schedule* and the table of the database is *class* (*teacherName*, *dateofclass*, *timeofclass*, *classroom*), which have been created by you using SQL command using the username *student*. The following command would be needed by you to create an active connection:

```
Connection connection = DriverManager.getConnection
```

```
("jdbc:mysql://localhost:3306/class_schedule", "student", "dbms");
```

You may have to perform several other commands, too. You may refer to further readings for more details.

Creating a form for data input and inserting data into the database

You may need to create a form using HTML or any scripting language and use GET or POST methods to transfer the data to the web server, which may execute the commands for inserting data into the database (only after establishing a connection with the database).

For example, the following commands will help you insert the data into the database:

```
String sqlinsert = "insert into class_schedule values (?, ?, ?, ?)";
PreparedStatement preparedstat = connection.prepareStatement (sqlinsert);
preparedstat.setString(1, teacherName);
preparedstat.setString(2, dateofclass);
preparedstat.setString(3, timeofclass);
preparedstat.setString(4, classroom);
```



```
prep.executeUpdate();
    preparedstat.close();
```

The commands shown above are just to demonstrate the programming. Please note that the variable name *connection* is the same as the variable name used while establishing the connection. The insert statement consists of four '?', filled up by the data obtained from the HTML form using the *setString* functions.

Accessing data from the database and displaying it as a webpage

For this purpose, you may create a query to access the database (after establishing a connection). The data obtained from the query may be put in a variable, which can be accessed record by record or complete data at a time to create the required webpage.

For example, you may display the content of all the classes using the following set of commands. This command may result in a number of rows of information being created by the output *System.out.println* function call.

```
String sqlselect = "select * from class_schedule ";
Statement statement = connection.createStatement();
ResultSet results=statement.executeQuery (sqlselect);
while(results.next())
    System.out.println(results.getString(1)+" "+rs.getString(2)+" "+
        rs.getString(3)+" "+rs.getString(4));
```

Many more statements are required to implement a web database. You may refer to the documentation of the web tools, languages and databases you use to implement a web application. While using the connection, the term *jdbc* was used. Let us discuss such terms in more detail.

Open Database Connectivity (ODBC) and JAVA Database Connectivity (JDBC)

There are many commercial DBMSs like DB2 (IBM), Oracle and MySQL (Oracle), SQL Server, MS-Access (Microsoft), PostgreSQL (Open Source) etc. Each DBMS has its own interface for programming (called Application Programming Interface or, in short, API) and data storage and indexing mechanisms. ODBC is a standard interface (API) that allows you to connect to a database of any DBMS and manipulate data using SQL commands. If you are using JAVA to write the interface functions to a DBMS, then you require JDBC. These interfaces provide a set of drivers that allow you to connect and access data from the database.

Closing of database connection: In general, it is recommended to close the connection after the required database operation has been performed.

Check Your Progress 3

- 1) What are the characteristics of multimedia data?

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- 2) List four application areas of multimedia databases.

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3) What are the steps required to access a database from a web application?

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4. Why is there a need for ODBC? Why do you need JDBC?

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16.6 SUMMARY

This Unit introduced you to some of the advanced technologies. The unit discusses the concepts relating to the distributed database management systems (DDBMS). A DDBMS replicates the fragmented data on various database sites. This unit explains the concepts of horizontal and vertical fragmentation of data. A query in distributed database may be submitted at any site of the database. This unit introduces how a distributed query will be processed. The unit also explains the triggering events and dynamic actions in the context of active database with the help of an example. XML is one of the popular ways of representing semi-structure data in a very simple format. This unit explained the data representation and validation in XML. The unit also presented a brief introduction to blockchain technology. The focus in the unit was to introduce you to basic concepts with the help of an example. Finally, the unit presents a brief introduction to multimedia database and web databases.

16.7 SOLUTIONS / ANSWERS

Check Your Progress 1

- 1) A DDBMS allows transparency to a user about the location of the data, which may be distributed among various sites of the database. DDBMS makes sure that data is made available to the user (if she has the access rights to use the data) even if one site is down. This is possible because DDBMS keeps track of the data replications. In addition, DDBMS is responsible for concurrent transaction management in distributed settings. The RDBMS, on the other hand, does not distribute data and, therefore, is less complex than DDBMS.
- 2) In such a case, you may only require the following fragment:

```
SELECT EnrNo, Name  
FROM Student  
WHERE Programme = 'PGDCA'
```
- 3) An active database is a database that has a set of active actions that are to be performed upon the occurrence of an event. Some of these triggering events can be the addition or deletion of a record or modification of a record.

Check Your Progress 2

- 1) The following is the XML document:

```
<studentMarks>
  <student>
    <rollno>12345</rollno>
    <name>Aman</name>
    <marksinsubject>
      <subject>Mathematics</subject>
      <marks>75</marks>
      <subject>Physics</subject>
      <marks>85</marks>
    </marksinsubject>
  </student>
  <student>
    <rollno>54321</rollno>
    <name>Arvin</name>
    <marksinsubject>
      <subject>Mathematics</subject>
      <marks>65</marks>
    </marksinsubject>
  </student>
</studentMarks>
```

The DTD would be as follows:

```
<!DOCTYPE studentMarks [
  <!ELEMENT (student (rollno, name, marksinsubject+))+>
  <!ELEMENT marksinsubject ( (subject, marks)+)>
  <!ELEMENT rollno ( #PCDATA )>
  <!ELEMENT name ( #PCDATA )>
  <!ELEMENT subject ( #PCDATA )>
  <!ELEMENT marks ( #PCDATA )>
]>
```

- 2) The nonce means the number used once. It is a random number appended to the blockchain content to ensure that no two blocks have the same hash value.
- 3) The change in the Block 3 E to F transaction to 100 will change the hash value of Block 3; this will no longer match the hash value of the previous block stored in Block 4. Further, the consensus protocol will not accept values from this block for any purpose.

Check Your Progress 3

- 1) The following are the basic characteristics of multimedia data:
- Textual Data:* includes long strings such as textual articles, can include multilingual UNICODE formats, and requires a keyword-based search.
 - Image Data:* large data requiring compression, use formats like GIF, PNG, JPEG, etc. images can be segmented and tagged for the purpose of similarity
 - Video and Animation Data:* has a large sequence of frames, video segments may be created, and popular compression formats are MPEG, AVI, etc.
 - Audio data:* may be identified for similarity of voice.
- 2) Medical databases
Bioinformatics

Multimedia for Home

3) The following steps are required to create a web database:

- You should have a valid database with proper records on a database server.
 - Create a connection string using the URL and port number of the database server. This sting should also include a valid username and password.
 - Open the connection, query the database using the parameters submitted by the user, and generate the results.
 - Format and display the results on the client, which may be a browser window.
4. ODBC helps in accessing the contents of the database using SQL commands, simplifying the programming of the databases. JDBC is an application programming interface that allows Java programmers to access any database.

